DATACO GLOBAL SUPPLY CHAIN ANALYTICS

End-to-End Data Analytics & Business Intelligence Project

Professional Portfolio Report

Excel • Python • SQL • Power BI

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1. Executive Summary

This project analyzes the DataCo Supply Chain dataset across 2015–2018 using Python for data preparation and EDA, MySQL for business analysis, and Power BI for interactive reporting. The project evaluates delivery performance, shipping modes, markets, customer segments, product profitability and operational risk.

Metric	Value
Total orders	180,519
Total revenue	$36.78M
Total profit	$3.97M
Late delivery rate	54.83%
On-time rate	17.84%
Largest shipping-mode share	Standard Class — 59.69%

Primary finding: late-delivery risk is the central operational issue. The final shipping-mode analysis identifies First Class as the highest-risk mode at 95.32%, making fulfillment-process investigation a priority.

- Standard Class carries the largest order volume, so improvements in this mode can affect a large number of orders.
- First Class shows the highest late-delivery rate in the final analytical report.
- The Power BI deliverable is organized into Executive Overview, Shipping Performance and Root Cause Drill-Down.
- The project connects descriptive analysis with business recommendations rather than stopping at charts.

2. Project Overview & Objectives

Business problem: management needs to understand where delivery delays occur and how shipping mode, market, customer segment, product category and profitability relate to overall supply-chain performance.

- Identify delivery-risk patterns by shipping mode, market, category and time.
- Quantify operational and financial implications of delivery failures.
- Use SQL aggregation to answer focused business questions.
- Build an interactive Power BI dashboard suite for different stakeholder needs.
- Translate analytical findings into practical business recommendations.

Dataset profile

Attribute	Value
Source	DataCo Supply Chain Dataset
Records	180,519 orders
Period	2015–2018
Markets	LATAM, Europe, Pacific Asia, USCA, Africa
Shipping modes	Standard Class, Second Class, First Class, Same Day

Technology stack

Tool	Role
Excel	Data exploration, Pivot Tables and supplementary analysis
Python / Pandas / NumPy	Cleaning, transformation, validation and analysis
Matplotlib / Seaborn	Exploratory visualizations
MySQL	Database analysis, aggregation and business queries
Power BI	Interactive dashboards, DAX, slicers and drill-down analysis

3. Data Cleaning & Preparation

- Removed Product Description, Order Zipcode, Customer Email, Customer Password, Product Image and Customer Street because they were not required for the analytical workflow.
- Filled missing Customer Lname values with “Unknown”.
- Converted order and shipping date fields to datetime format using mixed-format parsing.
- Validated the cleaned output before downstream SQL and Power BI analysis.

Representative Python workflow

```
import pandas as pd

df = pd.read_excel("DataCoSupplyChainDataset.PROJECT.xlsx")

df = df.drop(columns=[
    "Product Description", "Order Zipcode",
    "Customer Email", "Customer Password",
    "Product Image", "Customer Street"
])

df["Customer Lname"] = df["Customer Lname"].fillna("Unknown")
df["order date (DateOrders)"] = pd.to_datetime(
    df["order date (DateOrders)"], format="mixed", errors="coerce"
)
df["shipping date (DateOrders)"] = pd.to_datetime(
    df["shipping date (DateOrders)"], format="mixed", errors="coerce"
)
df.to_excel("DataCoSupplyChain_Clean.xlsx", index=False)
```

Data-quality outcome: the project report records zero duplicate rows and a validated cleaning workflow for downstream analysis.

4. Exploratory Data Analysis

EDA was used to understand delivery outcomes, shipping-mode volume, sales distribution and market concentration before SQL and Power BI analysis.

Delivery status

Outcome	Orders	Share
Late delivery	98,977	54.83%
Advance shipping	41,592	23.04%
Shipping on time	32,196	17.84%
Shipping canceled	7,754	4.30%

Shipping-mode volume

Mode	Orders	Share
Standard Class	107,752	59.69%
Second Class	35,216	19.51%
First Class	27,814	15.41%
Same Day	9,737	5.39%

EDA insight: Standard Class is the volume leader, while the delivery-risk analysis highlights premium modes as a performance-risk area. This distinguishes volume exposure from performance exposure.

5. SQL Database Analysis

The cleaned dataset was loaded into MySQL for structured business analysis. The project used COUNT, SUM, AVG, ROUND, GROUP BY, HAVING, ORDER BY, YEAR, MONTH and LIMIT.

Shipping-mode late-delivery query

```
SELECT `Shipping Mode`,
       COUNT(*) AS Total_Orders,
       SUM(`Late_delivery_risk`) AS Late_Orders,
       ROUND(AVG(`Late_delivery_risk`) * 100, 1) AS Late_Pct
FROM datacosupplychain_clean_sql
GROUP BY `Shipping Mode`
ORDER BY Late_Pct DESC;
```

Mode	Orders	Late orders	Late rate
First Class	27,814	26,513	95.32%
Second Class	35,216	26,987	76.63%
Same Day	9,737	4,454	45.74%
Standard Class	107,752	41,023	38.07%

Additional SQL analyses covered market profitability, negative-profit product categories, top customers by revenue and monthly profit trends.

6. Power BI Dashboard Suite

The Power BI deliverable was organized into three stakeholder-focused pages using a consistent teal-blue theme, KPI cards, slicers, comparative charts, time-series visuals and drill-down analysis.

Page	Audience	Main purpose
Executive Overview	Management / C-level	Revenue, profit, orders and delivery-risk overview
Shipping Performance	Logistics / Operations	Shipping delay, late orders, mode and market performance
Root Cause Drill-Down	Supply-chain managers	Geographic and shipping-mode investigation

Executive Overview

- Slicers: Year, Market, Customer Segment.
- KPIs: Total Revenue, Late Delivery %, Total Profit, Total Orders.
- Visuals: Sales by Market, trends, shipping-mode performance, profit by category, customer segments and delivery status.

Shipping Performance

- Slicers: date/year, Market and Shipping Mode.
- KPIs: Total Orders, Late Orders, Average Shipping Delay, Late Delivery %.
- Visuals: average delay by market/mode/category, time trend and shipping-mode distribution.

Root Cause Drill-Down

- Market-focused filtering and delivery KPIs.
- Geographic late-delivery analysis by country.
- Late orders by market, shipping-mode risk and market-versus-mode matrix.

7. Key Insights & Findings

Late-delivery crisis

54.83% of project-level orders are classified as late-delivery risk, making delivery performance the main operational concern.

First Class performance

First Class has the highest late-delivery rate in the final shipping-mode analysis at 95.32%.

Standard Class volume exposure

Standard Class represents 59.69% of orders, making it the largest volume lever for improvement.

Geographic risk

The root-cause dashboard shows very high late-delivery percentages across the displayed markets, indicating a broad operational issue.

Customer concentration

Top customers contribute a disproportionate share of revenue, creating retention and concentration risk.

Seasonality

Monthly profit analysis shows recurring variation that is relevant to inventory, staffing and capacity planning.

8. Recommendations & Action Plan

Priority	Recommended action	Horizon
Critical	Investigate First Class operations: carrier performance, processing time, handoffs, routes and service commitments.	Immediate
High	Trace fulfillment from order receipt through picking, packing and shipment to identify upstream bottlenecks.	30 days
High	Set delivery SLAs and monitor on-time performance by shipping mode, market and category.	Ongoing
High	Investigate the very high late rates in premium shipping modes and test process changes.	60 days
Medium	Review negative-profit product categories using cost, discount, shipping and order-level profitability evidence.	60 days
Medium	Protect high-value customer relationships through service-reliability and retention monitoring.	Ongoing

9. Technical Skills Demonstrated

Area	Evidence
Excel	Data exploration, Pivot Tables and supplementary analysis
Python	Pandas cleaning, missing-value treatment, datetime conversion and EDA
Visualization	Matplotlib / Seaborn exploratory charts
SQL / MySQL	Database creation, aggregation, GROUP BY/HAVING, ranking and time-based analysis
Power BI	Three-page dashboard suite, DAX, slicers, KPI cards and drill-down analysis
Business analysis	Root-cause framing, KPI interpretation and recommendations
Data storytelling	Technical findings translated into stakeholder-focused insights

10. Conclusion

The DataCo Global Supply Chain Analytics project demonstrates an end-to-end analytics workflow from raw data preparation through business intelligence and recommendations. The strongest evidence points to delivery performance as the primary operational challenge, with First Class showing the most severe shipping-mode risk in the final analysis.

The project combines Python, SQL, Excel and Power BI into one coherent portfolio workflow. The resulting dashboards provide different levels of analysis—from executive KPIs to shipping performance and root-cause drill-down—making the project suitable for junior data analyst portfolio presentation.

Portfolio presentation checklist

- GitHub repository contains the DataCo project structure and Power BI deliverables.
- Three Power BI pages are represented: Executive Overview, Shipping Performance and Root Cause Drill-Down.
- Python, SQL and Excel work support the Power BI analysis.
- This report documents the methodology, findings, recommendations and technical skills used.

Prepared for portfolio use — August 2026